Adaptive Threshold Correction Strategy for Sensorless High-Speed Brushless DC Drives Considering Zero-Crossing-Point Deviation

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***Abstract*—This article investigates the influence of two nonideal factors on zero-crossing-point-detection-based sensorless control of high-speed brushless dc (BLDC) motor, i.e., asymmetric machine parameters and the resistance tolerance of back electromotive force (EMF) measurement circuit. Influenced by the machine parameter asymmetry, the back EMF envelopes will shift horizontally from its ideal position. Furthermore, the acquisition accuracy of back EMF is limited by the resistance tolerance of measurement circuit. Inaccurate acquisition of back EMF will result in vertical deviation of the envelopes. These two nonideal factors decrease the commutation accuracy and deteriorate the control performance. In order to suppress the commutation errors, an adaptive threshold correction strategy, including horizontal and vertical corrections is proposed for eliminating the influence of these two nonideal factors, respectively. In addition, the proposed correction method does not depend on any parameters of the machine or the measurement circuit. Finally, the theoretical analysis and the proposed correction method are verified by experiment on a 54 000 r/min high-speed BLDC drive platform.**

***Index Terms*—Asymmetry, brushless dc (BLDC) motor, commutation correction, high speed, resistance tolerance, sensorless control.**

I. Introduction

RECENTLY, high-speed brushless dc (BLDC) motors have received much research attention due to high power density, high reliability, and small size for applications including air compressors, flywheels, and high-speed drilling spindles [1]–[4]. In a BLDC drive, six discrete rotor position signals are necessary and the Hall sensors are usually employed. The sensors not only increase the system cost and complexity but also are sensitive to the extreme environment, including high-temperature surrounding and electromagnetic interference. In addition, it is quite difficult to achieve high installation accuracy of Hall sensors, especially in small size motors. Therefore, various BLDC sensorless control techniques have been investigated over the last two decades.

BLDC sensorless schemes can be classified into back electromotive force (EMF) zero-crossing-point (ZCP) detection [5]–[11], back EMF integration [12], [13], and flux linkage functions [14]–[16]. The latter two methods rely on high sampling frequency ratio, which are not suited in a high-speed region. Besides, the ZCP detection of back EMF is easy to implement and requires minimal knowledge of machine parameters. Hence, for high-speed applications, the ZCP detection method is the most popular solution.

In order to improve the commutation accuracy, some nonideal factors have been investigated, which could be classified into two categories. The first category is related to the internal power angle between the current vector and the back EMF vector. It is reported that this internal power angle is influenced by winding inductance [17], [18] and nonideal back EMF [19], [20]. These literatures have shown that by setting appropriate commutation point, the internal power angle can be minimized and the control performance can be improved. The second category refers to the factors that can cause ZCP position errors. The most common reason is the phase delay caused by low-pass filters (LPFs) in the signal process of zero-crossing detection. In [21] and [22], the method to select suitable LPF cutoff frequency and the phase delay compensation was introduced. Besides, [23] described an issue of unbalanced ZCPs that could be caused by dc-link voltage drop and offset. The unbalanced ZCP detection will lead to current ripple and torque ripple. However, the research of the second category is not sufficient at present. Two other potential factors that will lead to ZCP deviation are rarely investigated, i.e., asymmetric machine parameters and the resistance tolerance of back EMF measurement circuit.

The effect of machine parameter asymmetry on sensorless field-oriented control was widely investigated in [24] and [25], whereas for the six-step control, this effect is rarely investigated except in [26]. However, this method can be only used for saliency BLDC motors, and bipolar pulsewidth modulation (PWM) is required. In practice, many factors will cause machine parameter asymmetry, including cabling, inverter, rough mechanical manufacturing, rotor eccentricity, or winding fault

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[24], [25]. Especially for the high-speed BLDC motor with low resistance and inductance, the parameters are very sensitive. Even the asymmetric PCB layout may lead to parameter asymmetry. This effect will have a negative influence on ZCP detection and deteriorate the control performance.

Besides, the resistance tolerance of back EMF measurement circuit is another ZCP deviation cause, which was ignored in most previous literatures. Basically, there are two methods to detect the ZCP. One is based on the voltage comparator [5]–[7]. In this method, the polar signal of back EMF can be obtained by the voltage comparator. Then, it will trigger the peripheral interrupt of the digital processor for commutation. The other ZCP detection method is PWM synchronization sampling by the analog-to-digital converter (ADC) [8]–[11]. The back EMF signal on the floating phase is sampled once or more during each PWM period. Then the ZCP detection is implemented in the algorithm. No matter which scheme is selected, the resistive voltage divider (RVD) is usually employed as the back EMF transducer for its simplicity and low cost. When considering the resistance tolerance, the resistance ratio will deviate from its nominal value due to temperature rise and aging effect. Especially for those applications in which the motor and drive boards are integrated within a compact space, the temperature will change dramatically during operation. The resistance ratio shift will cause ZCP deviation and further decrease the commutation accuracy.

In brief, the asymmetric machine parameters and the resistance tolerance of back EMF measurement circuit will lead to ZCP deviation and further deteriorate the drive performance. Moreover, the parameters of the machine and the measurement circuit are not constant. For every single motor and drive, the parameters may initially be different and may also change during operation. Therefore, a commutation correction method that is parameter independent is expected. In this article, an adaptive threshold correction scheme for ZCP deviation is proposed. First, the theoretical analyses of asymmetric machine parameters and the resistance tolerance are illustrated in Sections II and III, respectively. Then, the principle of the proposed adaptive threshold correction strategy is presented in Section IV. Finally, experimental validation is presented in Section V, and Section VI concludes this article.

II. Horizontal Deviation of Envelopes Caused by Machine Parameter Asymmetry

The drive topology of a high-speed motor is shown in Fig. 1(a). The BLDC motor can be modeled as

$$\begin{cases} v_{\mathrm{AG}} - v_{\mathrm{NG}} = R_{\mathrm{A}} i_{\mathrm{A}} + L_{\mathrm{A}} \frac{d}{dt} i_{\mathrm{A}} + e_{\mathrm{A}} \\ v_{\mathrm{BG}} - v_{\mathrm{NG}} = R_{\mathrm{B}} i_{\mathrm{B}} + L_{\mathrm{B}} \frac{d}{dt} i_{\mathrm{B}} + e_{\mathrm{B}} \\ v_{\mathrm{CG}} - v_{\mathrm{NG}} = R_{\mathrm{C}} i_{\mathrm{C}} + L_{\mathrm{C}} \frac{d}{dt} i_{\mathrm{C}} + e_{\mathrm{C}} \end{cases} \tag{1}$$

where v_{XG} is the terminal voltage, v_{NG} is the machine neutral voltage, i_{X} is the phase current, e_{X} is the phase back EMF, and R_{X} is the stator resistance. L_{X} is the stator inductance combining self-inductance L'_{X} and mutual inductance M, i.e., $L_{\mathrm{X}} = L'_{\mathrm{X}} - M$. The subscript "X" denotes phase A, B, or C. Here the resistances and self-inductances in three phases are assumed unbalanced. According to previous documents [17], high-speed BLDC motors have sinusoidal phase back EMFs as a result of two-pole diametrically magnetized magnet. Thus, the back EMFs can be expressed as

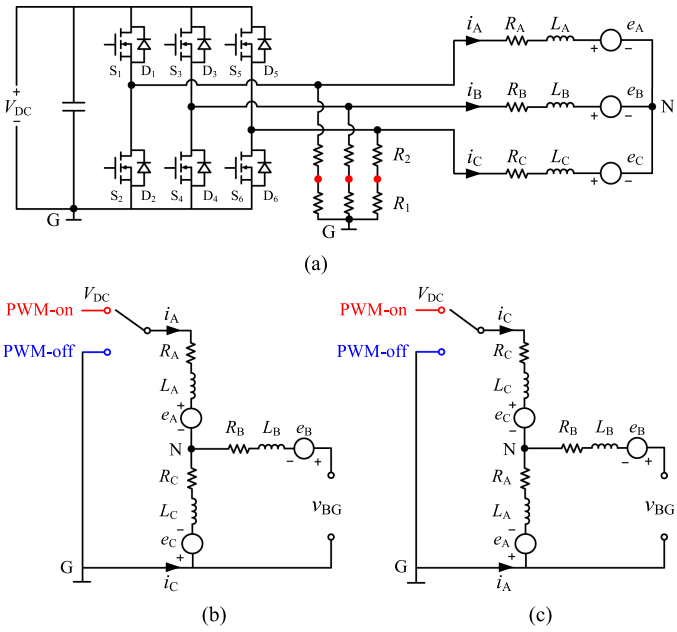


Fig. 1. Drive topology of high-speed BLDC motor. (a) Overall drive circuit. (b) Equivalent circuit in sector II. (c) Equivalent circuit in sector V.

$$\begin{bmatrix} e_{\mathrm{A}} \\ e_{\mathrm{B}} \\ e_{\mathrm{C}} \end{bmatrix} = \lambda\omega \begin{bmatrix} \sin(\theta) \\ \sin\left(\theta - \frac{2}{3}\pi\right) \\ \sin\left(\theta + \frac{2}{3}\pi\right) \end{bmatrix} \tag{2}$$

where λ, ω, and θ are the flux linkage, electrical speed, and rotor position, respectively.

For the six-step commutation scheme, an electrical cycle is divided into six sectors as shown. In each 60° sector, only two of the three phases are energized. The floating phase contains the back EMF information, which can be used for commutation detection. In position sensorless BLDC motor drives, the commutation signal can be obtained by lagging 30° after the ZCP of back EMF.

In this section, the influence of machine parameter asymmetry on ZCP detection will be analyzed. Two opposite sectors, II and V as shown in Fig. 1(b) and (c) and Fig. 2 are illustrated first and other sectors can be derived.

The BLDC model (1) can be transformed into Laplace domain as

$$\begin{cases} V_{\mathrm{AG}}(s) - V_{\mathrm{NG}}(s) = (R_{\mathrm{A}} + sL_{\mathrm{A}}) \cdot I_{\mathrm{A}}(s) + E_{\mathrm{A}}(s) \\ V_{\mathrm{BG}}(s) - V_{\mathrm{NG}}(s) = (R_{\mathrm{B}} + sL_{\mathrm{B}}) \cdot I_{\mathrm{B}}(s) + E_{\mathrm{B}}(s) \\ V_{\mathrm{CG}}(s) - V_{\mathrm{NG}}(s) = (R_{\mathrm{C}} + sL_{\mathrm{C}}) \cdot I_{\mathrm{C}}(s) + E_{\mathrm{C}}(s) \end{cases} \tag{3}$$

where s denotes the Laplace operator. In sectors II and V, phases A and C are conducting and phase B is floating, it yields $I_{\mathrm{A}}(s) + I_{\mathrm{C}}(s) = 0$ and $I_{\mathrm{B}}(s) = 0$. The terminal voltage $V_{\mathrm{BG}}(s)$ can be

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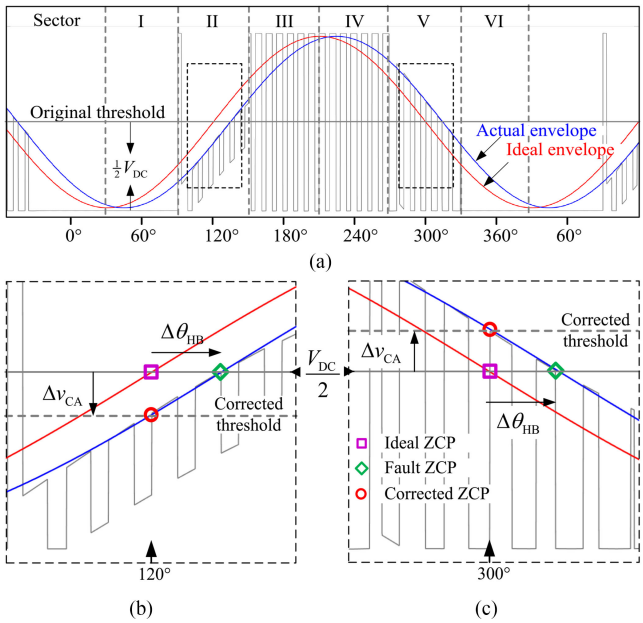


Fig. 2. Schematic diagram of terminal voltage in Phase B considering machine parameter asymmetry. (a) Terminal voltage, ideal, and actual envelopes. (b) Zoomed window for Sector II. (c) Zoomed window for Sector V.

calculated as

$$V_{BG}(s) = V_{NG}(s) + E_B(s) = \frac{1}{2}\left(V_{AG}(s) + V_{CG}(s)\right) + \frac{3}{2}E_B(s) - \frac{1}{2}\Delta V_H(s) \tag{4}$$

$$\Delta V_H(s) = \left[V_{AG}(s) - V_{CG}(s) - \left(E_A(s) - E_C(s)\right)\right] \times \frac{\Delta R_{AC} + s\Delta L_{AC}}{\Sigma R_{AC} + s\Sigma L_{AC}} \tag{5}$$

where $\Delta V_H(s)$ is related to asymmetric parameters, and $\Delta R_{AC} = R_A - R_C$, $\Delta L_{AC} = L_A - L_C$, $\Sigma R_{AC} = R_A + R_C$, and $\sum L_{AC} = L_A + L_C$. Then, transform $V_{BG}(s)$ back to time domain, it yields

$$v_{BG}(t) = \mathcal{L}^{-1}\left[V_{BG}(s)\right] = \frac{1}{2}\left(v_{AG}(t) + v_{CG}(t)\right) + \frac{3}{2}e_B(t) - \frac{1}{2}\mathcal{L}^{-1}\left[\Delta V_H(s)\right]. \tag{6}$$

The additional term $-\frac{1}{2}\mathcal{L}^{-1}[\Delta V_H(s)]$ is the voltage shift caused by asymmetric parameters. The terminal voltage $v_{BG}(t)$ will be calculated in sectors II and V, respectively.

A. Terminal Voltage in Sector II

During sector II as shown in Fig. 1(b), phases A and C are energized and phase B is floating. This floating phase B containing the back EMF signal information can be used for commutation detection. Power device S_1 is controlled by chopper signal and S_6 keeps conducting in this sector. The circuit switches between PWM-ON and PWM-OFF states as follows.

PWM-ON: phases A and C are connected to positive and negative dc-link rails through S_1 and S_6, respectively.

PWM-OFF: due to the freewheeling effect, both two phases are connected to the negative dc-link rail with D_2 and S_6.

The back EMF envelope can be extracted by sampling at either PWM-ON or PWM-OFF state [9]–[11]. As the motor speed and PWM duty ratio increase, the PWM-ON duration increases and the PWM-OFF duration decreases. To guarantee the detection of ZCP at a high-speed region, this proposed method is based on back EMF sampling at PWM-ON time. During this PWM-ON period, $v_{AG} = V_{DC}$, $v_{CG} = 0$, and $\mathcal{L}^{-1}[\Delta V_H(s)]$ can be calculated in Appendix A. Then, the terminal voltage envelope of the floating phase $v_{BG}(t)$ in sector II can be expressed as

$$v_{BG}(t) = \frac{1}{2}V_{DC} + \frac{3}{2}e_B(t) + \Delta v_{AC} \tag{7}$$

where the voltage shift Δv_{AC} is

$$\Delta v_{AC} = -\frac{1}{2}(V_{DC} - \sqrt{3}\lambda\omega)\frac{L_A - L_C}{L_A + L_C}. \tag{8}$$

As shown in Fig. 2(b), with symmetrical parameters, the dc offset of terminal voltage envelope is $\frac{1}{2}V_{DC}$. However, when the machine parameter asymmetry is taken into account, the additional term Δv_{AC} moves the terminal voltage envelope, as shown in Fig. 2(b). If $\frac{1}{2}V_{DC}$ is still selected as the zero-crossing threshold, an angle error will be definitely caused.

B. Terminal Voltage in Sector V

Opposite to sector II, as shown in Fig. 1(c), S_5 is controlled by chopper signal and S_2 keeps conducting in sector V. The circuit switches between PWM-ON and PWM-OFF states as follows.

PWM-ON: phases A and C are connected to negative and positive dc-link rails through S_2 and S_5, respectively.

PWM-OFF: due to the freewheeling effect, both two phases are connected to the negative dc-link rail with S_2 and D_6.

During PWM-ON period, $v_{AG} = 0$, $v_{CG} = V_{DC}$, and $\mathcal{L}^{-1}[\Delta V_H(s)]$ is calculated in Appendix A. The terminal voltage envelope of the floating phase $v_{BG}(t)$ in sector V can be expressed as

$$v_{BG}(t) = \frac{1}{2}V_{DC} + \frac{3}{2}e_B(t) + \Delta v_{CA} \tag{9}$$

where the voltage shift Δv_{CA} is

$$\Delta v_{CA} = -\frac{1}{2}(V_{DC} - \sqrt{3}\lambda\omega)\frac{L_C - L_A}{L_C + L_A}. \tag{10}$$

It is shown in Fig. 2(c), the terminal voltage envelope moves by Δv_{CA}. From (7) and (9), it can be found the voltage errors Δv_{AC} and Δv_{CA} exist in the terminal voltage of the floating phase during sectors II and V, respectively. Besides, it yields $\Delta v_{AC} + \Delta v_{CA} = 0$, since the motor speed will not change dramatically in the steady-state operation.

C. Influence of Machine Parameter Asymmetry on ZCP Detection

The terminal voltage envelopes in sectors II and V have been analyzed. The results of other sectors can also be obtained in the similar way. Table I summaries voltage shifts, position errors,

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TABLE I
TERMINAL VOLTAGE ENVELOPE SHIFTS AND ZCP POSITION ERRORS

Sector	Mode	EMF (F/R)	Δv_{H}	Δv_{V}	$\Delta\theta$
I	A+B-C~	F	Δv_{AB}	Δv_{VC}	$\Delta\theta_{\mathrm{HC}}+\Delta\theta_{\mathrm{VC}}$
II	A+B~C-	R	$-\Delta v_{\mathrm{CA}}$	Δv_{VB}	$\Delta\theta_{\mathrm{HB}}-\Delta\theta_{\mathrm{VB}}$
III	A~B+C-	F	Δv_{BC}	Δv_{VA}	$\Delta\theta_{\mathrm{HA}}+\Delta\theta_{\mathrm{VA}}$
IV	A-B+C~	R	$-\Delta v_{\mathrm{AB}}$	Δv_{VC}	$\Delta\theta_{\mathrm{HC}}-\Delta\theta_{\mathrm{VC}}$
V	A-B~C+	F	Δv_{CA}	Δv_{VB}	$\Delta\theta_{\mathrm{HB}}+\Delta\theta_{\mathrm{VB}}$
VI	A~B-C+	R	$-\Delta v_{\mathrm{BC}}$	Δv_{VA}	$\Delta\theta_{\mathrm{HA}}-\Delta\theta_{\mathrm{VA}}$

and the back EMF direction of the floating phase, i.e., rising or falling. We define the voltage shift caused by parameter asymmetry as

$$\Delta v_{\mathrm{XY}}=-\frac{1}{2}(V_{\mathrm{DC}}-\sqrt{3}\lambda\omega)\cdot\frac{L_{\mathrm{X}}-L_{\mathrm{Y}}}{L_{\mathrm{X}}+L_{\mathrm{Y}}}. \tag{11}$$

From the above analysis, asymmetric machine parameters result in the movement of the terminal voltage envelope in the floating phase. As shown in Fig. 2, the envelope moves down by Δv_{CA} in sector II and moves up by Δv_{CA} in sector V. In the vertical direction, the envelope moves oppositely in sectors II and V. From another point of view, both the vertical movements in sectors II and V can be unified into a horizontal movement of $\Delta\theta_{\mathrm{HB}}$. Hence, influenced by the machine parameter asymmetry, the terminal voltage envelopes move horizontally, leading to ZCP deviations.

The envelope around each ZCP can be regarded as a straight line with a slope of $\frac{3}{2}\omega\lambda$. The voltage shifts have a linear relationship with ZCP angle errors as

$$[\Delta v_{\mathrm{AB}}\ \Delta v_{\mathrm{BC}}\ \Delta v_{\mathrm{CA}}]=\frac{3}{2}\omega\lambda\cdot[\Delta\theta_{\mathrm{HC}}\ \Delta\theta_{\mathrm{HA}}\ \Delta\theta_{\mathrm{HB}}]. \tag{12}$$

Besides, the sum of three voltage shifts can be calculated as

$$\begin{aligned}&\Delta v_{\mathrm{AB}}+\Delta v_{\mathrm{BC}}+\Delta v_{\mathrm{CA}}\\&=-\frac{1}{2}\left(V_{\mathrm{DC}}-\sqrt{3}\lambda\omega\right)\left(\tfrac{L_{\mathrm{A}}-L_{\mathrm{B}}}{L_{\mathrm{A}}+L_{\mathrm{B}}}+\tfrac{L_{\mathrm{B}}-L_{\mathrm{C}}}{L_{\mathrm{B}}+L_{\mathrm{C}}}+\tfrac{L_{\mathrm{C}}-L_{\mathrm{A}}}{L_{\mathrm{C}}+L_{\mathrm{A}}}\right)\\&=\frac{1}{2}\left(V_{\mathrm{DC}}-\sqrt{3}\lambda\omega\right)\cdot\underbrace{\frac{L_{\mathrm{A}}-L_{\mathrm{B}}}{L_{\mathrm{A}}+L_{\mathrm{B}}}}_{\Delta L_{\mathrm{AB}}}\cdot\underbrace{\frac{L_{\mathrm{B}}-L_{\mathrm{C}}}{L_{\mathrm{B}}+L_{\mathrm{C}}}}_{\Delta L_{\mathrm{BC}}}\cdot\underbrace{\frac{L_{\mathrm{C}}-L_{\mathrm{A}}}{L_{\mathrm{C}}+L_{\mathrm{A}}}}_{\Delta L_{\mathrm{CA}}}.\end{aligned} \tag{13}$$

It should be noted that three inductance related terms, ΔL_{AB}, ΔL_{BC}, and ΔL_{CA} are all very small. Then, their product is actually close to zero. Thus, (13) can be written as

$$\Delta v_{\mathrm{AB}}+\Delta v_{\mathrm{BC}}+\Delta v_{\mathrm{CA}}\approx 0. \tag{14}$$

Additionally, due to the linear relation of (12), the sum of three ZCP angle errors is also approximately zero:

$$\Delta\theta_{\mathrm{HA}}+\Delta\theta_{\mathrm{HB}}+\Delta\theta_{\mathrm{HC}}\approx 0. \tag{15}$$

The commutation instant is obtained by lagging 30° after every ZCP. Without an accurate ZCP, the commutation instant will not be identified. Therefore, machine parameter asymmetry has a negative impact on position sensorless BLDC drives.

III. VERTICAL DEVIATION OF ENVELOPES CAUSED BY RESISTANCE TOLERANCE OF MEASUREMENT CIRCUIT

The resistance tolerance of measurement circuit is another ZCP deviation cause. The maximum terminal voltage reaches dc-link level. However, the on-chip analog-to-digital (A-D) converter can only sample low-voltage signal, e.g., 0–3 V. Thus, the RVD is usually employed as the transducer for terminal voltage sampling. The resistance precision will definitely affect the commutation accuracy. In this section, the commutation error caused by resistance tolerance is analyzed.

As shown in Fig. 1, the A-D converter samples the terminal voltage signal on the low-side resistor of RVD. The scale factor of the resistance can be designed as

$$\frac{R_1}{R_1+R_2}=\frac{V_{\mathrm{AD}}}{V_{\mathrm{DC}}} \tag{16}$$

where V_{AD} denotes the maximum sampling voltage, and R_1 and R_2 are the low- and high-side resistance of RVD, respectively. In practice, the actual resistance has a slight difference from the nominal value. Hence, the equivalent gain K_{R} is

$$K_{\mathrm{R}}=\frac{R'_1}{R'_1+R'_2}\cdot\frac{R_1+R_2}{R_1} \tag{17}$$

where superscript "′" denotes the value taking resistance tolerance into account. Then, the acquired phase B terminal voltage can be expressed as

$$v'_{\mathrm{BG}}=v_{\mathrm{BG}}\cdot K_{\mathrm{R}}=\frac{1}{2}K_{\mathrm{R}}V_{\mathrm{dc}}+\frac{3}{2}K_{\mathrm{R}}e_B. \tag{18}$$

As shown here, the original dc offset $\frac{1}{2}V_{\mathrm{DC}}$ is shifted to $\frac{1}{2}K_{\mathrm{R}}V_{\mathrm{DC}}$ due to the resistance tolerance of RVD. The voltage shift can be expressed as

$$\Delta v_{\mathrm{VB}}=\frac{1}{2}\left(K_{\mathrm{R}}-1\right)V_{\mathrm{DC}}. \tag{19}$$

Similar to the analysis method in (12), the ZCP angle errors and voltage shifts given in Table I have linear relation as

$$[\Delta v_{\mathrm{VA}}\ \Delta v_{\mathrm{VB}}\ \Delta v_{\mathrm{VC}}]=\frac{3}{2}\omega\lambda\,[\Delta\theta_{\mathrm{VA}}\ \Delta\theta_{\mathrm{VB}}\ \Delta\theta_{\mathrm{VC}}]. \tag{20}$$

Therefore, as the resistance value deviates from the nominal one, the voltage error increases. Let us consider an extreme situation, R'_1 and R'_2 have upper and lower limit values, i.e., $R'_1=(1+\sigma)R_1$ and $R'_2=(1-\sigma)R_2$, where σ is the tolerance ratio. Finally, the ZCP angle error can be calculated as

$$\Delta\theta_{\mathrm{VB}}=\frac{1}{\sqrt{3}\omega_{\mathrm{pu}}}\cdot\left(\frac{1+\sigma}{(1+\sigma)+(1-\sigma)\left(\frac{V_{\mathrm{DC}}}{V_{\mathrm{AD}}}-1\right)}\cdot\frac{V_{\mathrm{DC}}}{V_{\mathrm{AD}}}-1\right) \tag{21}$$

where $\omega_{\mathrm{pu}}=\omega/\omega_{\mathrm{base}}$ denotes the per-unit value of speed, and the base-speed ω_{base} can be defined as $\sqrt{3}\lambda\omega_{\mathrm{base}}=V_{\mathrm{DC}}$.

Obviously, the voltage acquisition accuracy is limited by the resistance tolerance. The resistance will change with the temperature. Especially for those applications in which the motor and the drive boards are integrated within a compact space, the temperature will change significantly during operation. Besides, as the resistor gets old the resistance will change. For high-voltage

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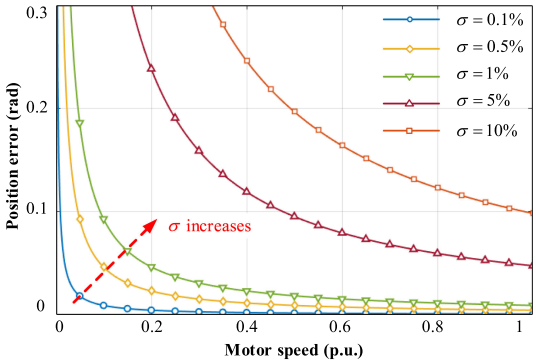


Fig. 3. Analysis result of ZCP angle error caused by vertical shift of terminal voltage. $V_{\mathrm{AD}} = 3$ V, $V_{\mathrm{DC}} = 15$ V.

application, due to voltage limit, more than two resistors are cascaded to reduce the voltage drop on each resistor. This situation is even more complicated. If the parameter design in the software algorithm is just based on the nominal resistance, it will definitely lead to commutation errors.

Fig. 3 shows the analysis results of the ZCP angle error. Some popular resistance tolerances from 0.1 to 10% are analyzed. As we can see, the ZCP angle error increases as the resistance precision decreases. High-precision resistors minimize the tolerance, the temperature shift, and aging, which means high commutation accuracy. But the cost will be high, which is the last thing the consumers want, especially in the household appliance market.

IV. Proposed Adaptive Threshold Correction Algorithm for Envelope Deviation

In this section, an adaptive threshold correction scheme is proposed. Fig. 4 illustrates the implementation of the overall scheme. The terminal voltage signals are sampled with RVD by the on-chip ADC. Then an adaptive threshold correction loop is utilized. In this algorithm, six independent zero-crossing thresholds are used for ZCP detection. These six thresholds are adjusted during operation in real time. Two CPU timers are utilized. Timer A is for time interval measurement of ZCPs. Six latest measurement results of ZCP time intervals, denoted as $t_{12} \ldots t_{61}$, are stored in a buffer. Timer B is used to trigger commutation 30° after each ZCP. In order to improve the accuracy of time interval measurement, the sampling time-delay compensation method is adopted.

A. Vertical Correction for Resistance Tolerance of Measurement Circuit

Fig. 5 shows the vertical shift of back EMF envelope caused by resistance tolerance of measurement circuit. Fig. 6 shows the zoomed area of the triangles in Fig. 5. The dashed and solid lines demote the back EMF envelopes before and after shift, respectively. It is assumed that $K_{\mathrm{R}} > 1$ and the back EMF envelope moves upward. Ideally, the phase B ZCPs should locate at the intersection of the phase B envelope e_{B} and the half dc-link voltage $\frac{1}{2}V_{\mathrm{DC}}$, i.e., 120° in sector II and 300° in sector V. However, when the resistance ratio changes, the envelope moves upward by Δv_{VB} to $\hat{e}_{\mathrm{B}}$, and a fault ZCP will be detected with the original threshold $\frac{1}{2}V_{\mathrm{DC}}$.

As shown in Fig. 6, the principle of the vertical correction is to superpose a common mode bias u_{VB} to $\frac{1}{2}V_{\mathrm{DC}}$, i.e., $\frac{1}{2}V_{\mathrm{DC}} + u_{\mathrm{VB}}$. This new threshold is adopted for both sectors II and V. The angle error $\varepsilon_{\mathrm{VB}}$ can be extracted in the real time. By controlling $\varepsilon_{\mathrm{VB}}$ to zero, the common mode bias u_{VB} will converge to Δv_{VB}. Then, even if the vertical shift exists, the ZCP will be corrected back to the ideal position, i.e., 120° and 300°.

As shown in Fig. 5, the phase-ZCP-interval $\hat{\theta}_{25}$ (II–V) and $\hat{\theta}_{52}$ (V–II) can be expressed as

$$\begin{cases} \hat{\theta}_{25} = \theta_{25} + 2\varepsilon_{\mathrm{VB}} \\ \hat{\theta}_{52} = \theta_{52} - 2\varepsilon_{\mathrm{VB}} \end{cases} \tag{22}$$

where the superscript "^" denotes measurement values. It is obvious that $\theta_{25} = \theta_{52} = \pi$ in (22), then the angle errors can be deduced as

$$\varepsilon_{\mathrm{VB}} = \frac{1}{4}(\hat{\theta}_{25} - \hat{\theta}_{52}). \tag{23}$$

In practice, $\hat{\theta}_{25}$ and $\hat{\theta}_{52}$ can be calculated by the CPU timer

$$\begin{cases} \hat{\theta}_{25} = \frac{t_{25}}{t_{\mathrm{period}}} \cdot 2\pi = \frac{t_{23}+t_{34}+t_{45}}{t_{\mathrm{period}}} \cdot 2\pi \\ \hat{\theta}_{52} = \frac{t_{52}}{t_{\mathrm{period}}} \cdot 2\pi = \frac{t_{56}+t_{61}+t_{12}}{t_{\mathrm{period}}} \cdot 2\pi \end{cases} \tag{24}$$

where t_{period} is the fundamental period and be expressed as

$$t_{\mathrm{period}} = t_{12} + t_{23} + \cdots + t_{61}. \tag{25}$$

Here, the correction process of phase B is taken for instance. The correction for the other two phases can also be achieved in the same way, which is shown in Fig. 7. The main principle is to search for a threshold, with which the upper- and lower-half-cycle have the same time duration. It should be noted that, precise resistance values are not required to be determined in advance during this process. Therefore, no matter how the resistance ratio of RVD changes, the suitable zero-crossing threshold can be searched.

B. Horizontal Correction for Machine Parameter Asymmetry

Fig. 8 shows the horizontal shift of back EMF envelope caused by machine parameter asymmetry. Fig. 9 shows the zoomed area of the triangles of phase B in Fig. 8. Influenced by asymmetric parameters, the envelope moves downward in sector II and moves upward in sector V, which can be unified into a right movement of $\Delta\theta_{\mathrm{HB}}$. Thus, fault ZCPs will be detected with the original threshold $\frac{1}{2}V_{\mathrm{DC}}$.

As shown in Fig. 9, the horizontal correction process is similar to the vertical correction. The difference is that differential-mode biases are used, which means $\frac{1}{2}V_{\mathrm{DC}} - u_{\mathrm{HB}}$ is used for sector II and $\frac{1}{2}V_{\mathrm{DC}} + u_{\mathrm{HB}}$ is used for sector V as new thresholds. Then, by controlling $\varepsilon_{\mathrm{HB}}$ to zero, the differential- mode bias u_{HB} will converge to Δv_{CA} and the ZCP will be corrected. When considering horizontal correction for all three phases, it should be noted that only two of the three terms u_{HA} and u_{HB} are required and the third one u_{HC} can be calculated directly from $u_{\mathrm{HA}} + u_{\mathrm{HB}} + u_{\mathrm{HC}} = 0$ as a result of (14).

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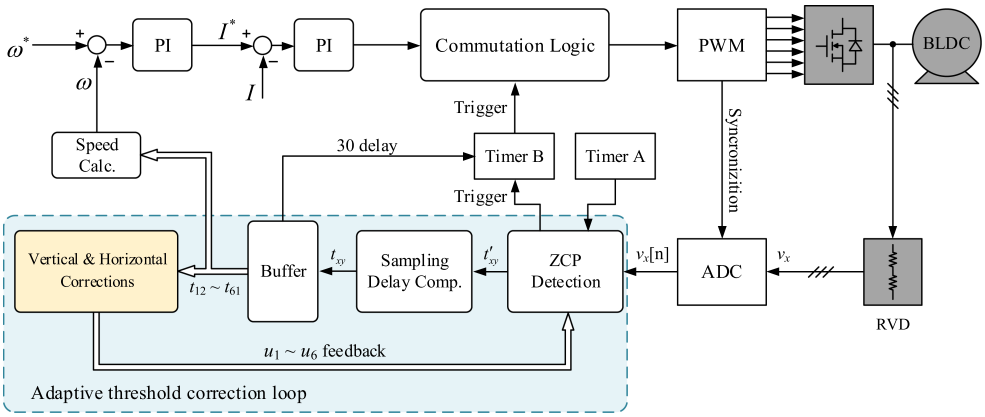


Fig. 4. Schematic diagram of sensorless control system with adaptive threshold correction strategy.

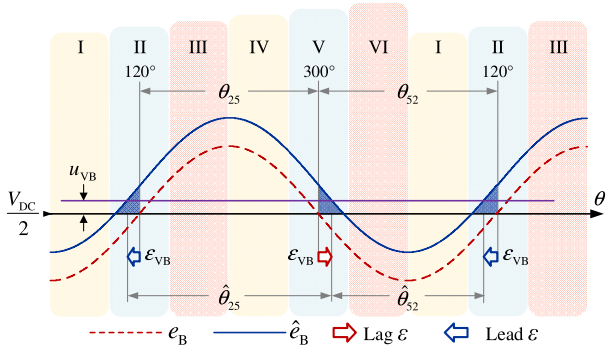


Fig. 5. Schematic diagram of vertical deviation of back EMF envelopes caused by resistance tolerance of measurement circuit.

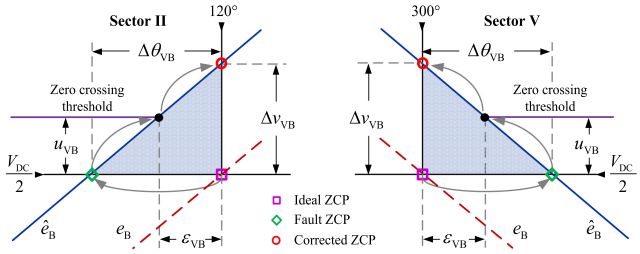


Fig. 6. Vertical correction process of zero-crossing threshold.

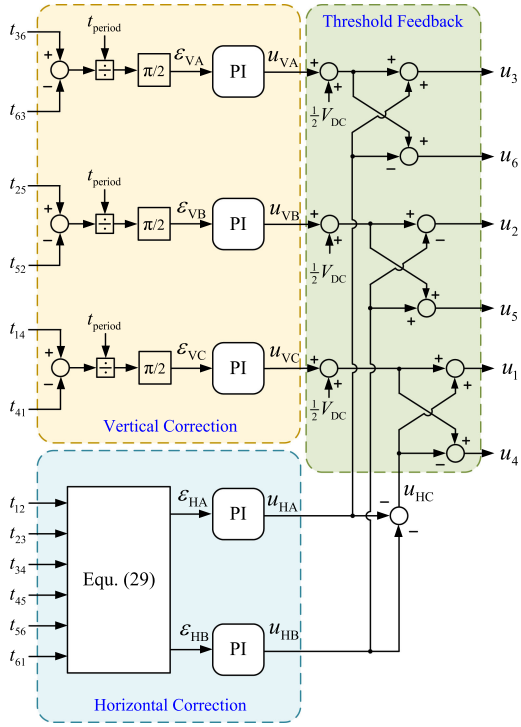


Fig. 7. Schematic diagram of adaptive threshold correction including vertical and horizontal corrections.

According to (15), the sum of three ZCP angle errors is approximately zero. Hence, there are two advanced phases and one lagged phase or one advanced phase and two lagged phases. In Fig. 8, phase A is assumed advanced and other two phases are assumed lagged. The sector ZCP intervals $\hat{\theta}_{12}\cdots\hat{\theta}_{61}$ can be modeled as

$$\begin{cases} \hat{\theta}_{12}(\hat{\theta}_{45}) = \theta_{12}\,(\theta_{45}) + \varepsilon_{\mathrm{HB}} - \varepsilon_{\mathrm{HC}} \\ \hat{\theta}_{23}(\hat{\theta}_{56}) = \theta_{23}\,(\theta_{56}) + \varepsilon_{\mathrm{HA}} - \varepsilon_{\mathrm{HB}} \\ \hat{\theta}_{34}(\hat{\theta}_{61}) = \theta_{34}\,(\theta_{61}) + \varepsilon_{\mathrm{HC}} - \varepsilon_{\mathrm{HA}} \end{cases}. \tag{26}$$

The term in the bracket indicates they have the same value, for instance $\hat{\theta}_{12} = \hat{\theta}_{45}$. Due to the linear relation, the following equation can be obtained:

$$\varepsilon_{\mathrm{HA}} + \varepsilon_{\mathrm{HB}} + \varepsilon_{\mathrm{HC}} = 0. \tag{27}$$

Also we have $\theta_{12} = \theta_{23} = \cdots = \theta_{61} = \pi/3$. Now, (27) and any two of three equations in (26) can be used to solve the equations to obtain the ZCP angle errors, it yields

$$\begin{bmatrix} \varepsilon_{\mathrm{HA}} \\ \varepsilon_{\mathrm{HB}} \end{bmatrix} = \frac{1}{3}\begin{bmatrix} 1 & 2 & 0 \\ 2 & 0 & 1 \end{bmatrix}\begin{bmatrix} \hat{\theta}_{12}(\hat{\theta}_{45}) \\ \hat{\theta}_{23}(\hat{\theta}_{56}) \\ \hat{\theta}_{34}(\hat{\theta}_{61}) \end{bmatrix} - \frac{\pi}{3}\begin{bmatrix} 1 \\ 1 \end{bmatrix}. \tag{28}$$

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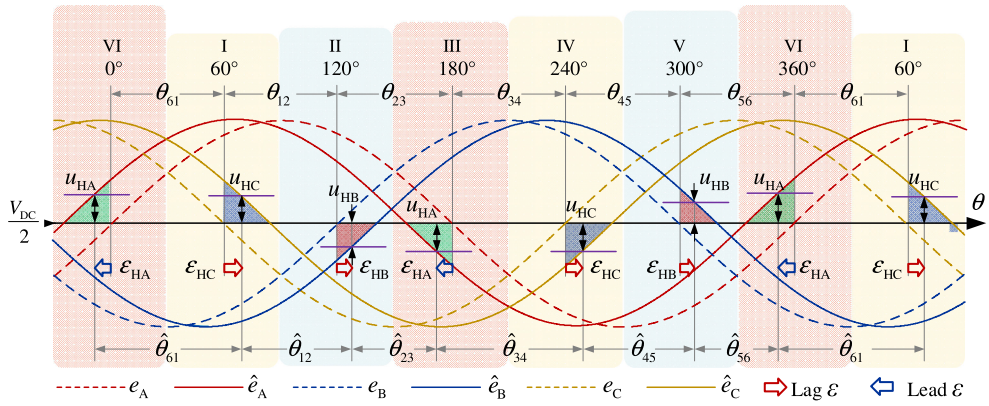


Fig. 8. Schematic diagram of horizontal deviation of back EMF envelopes caused by machine parameter asymmetry.

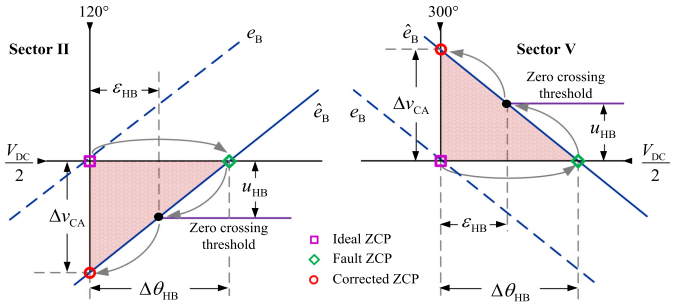


Fig. 9. Horizontal correction process of zero-crossing threshold.

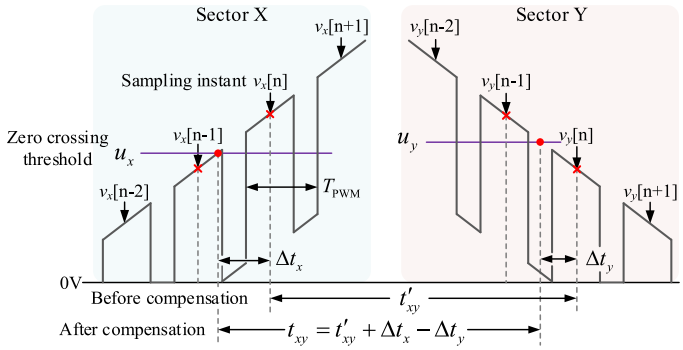


Fig. 10. Schematic diagram of sampling time-delay compensation, where $v_x[n]$ and $v_y[n]$ denote the nth terminal voltage samples in sectors X and Y, respectively; T_{PWM} denotes the PWM period; Δt_x and Δt_y denote the time delay in sectors X and Y, respectively; and t'_{xy} and t_{xy} denote raw and precise ZCP time intervals, respectively.

In practice, all the angle terms can be calculated by time interval signals obtained from the CPU timer as

$$\begin{bmatrix} \varepsilon_{\mathrm{HA}} \\ \varepsilon_{\mathrm{HB}} \end{bmatrix} = \frac{1}{3} \cdot \frac{2\pi}{t_{\mathrm{period}}} \cdot \begin{bmatrix} 1 & 2 & 0 \\ 2 & 0 & 1 \end{bmatrix} \begin{bmatrix} t_{12}\,(t_{45}) \\ t_{23}\,(t_{56}) \\ t_{34}\,(t_{61}) \end{bmatrix} - \frac{\pi}{3} \begin{bmatrix} 1 \\ 1 \end{bmatrix}. \tag{29}$$

Here, (29) gives the method to extract the angle errors $\varepsilon_{\mathrm{HA}}$ and $\varepsilon_{\mathrm{HB}}$. By controlling these angle errors to zero, differential-mode biases u_{HA}, u_{HB}, and u_{HC} will converge to Δv_{BC}, Δv_{AB}, and Δv_{CA}, respectively. Then the horizontal deviation caused by asymmetric parameters can be eliminated.

Finally, as shown in Fig. 7, three common-mode biases u_{VA}, u_{VB}, and u_{VC} can be obtained in the vertical correction; three differential-mode biases u_{HA}, u_{HB}, and u_{HC} can be obtained in the horizontal correction. Based on Table I, six zero-crossing thresholds can be calculated as

$$\begin{cases} u_3 = \frac{1}{2}V_{\mathrm{DC}} + u_{\mathrm{VA}} + u_{\mathrm{HA}} \\ u_6 = \frac{1}{2}V_{\mathrm{DC}} + u_{\mathrm{VA}} - u_{\mathrm{HA}} \end{cases} \tag{30}$$

$$\begin{cases} u_2 = \frac{1}{2}V_{\mathrm{DC}} + u_{\mathrm{VB}} - u_{\mathrm{HB}} \\ u_5 = \frac{1}{2}V_{\mathrm{DC}} + u_{\mathrm{VB}} + u_{\mathrm{HB}} \end{cases} \tag{31}$$

$$\begin{cases} u_1 = \frac{1}{2}V_{\mathrm{DC}} + u_{\mathrm{VC}} + u_{\mathrm{HC}} \\ u_4 = \frac{1}{2}V_{\mathrm{DC}} + u_{\mathrm{VC}} - u_{\mathrm{HC}} \end{cases}. \tag{32}$$

It should be noted that during the whole process, any parameters of the machine or the measurement circuit of back EMF are not required. Besides, the proposed adaptive correction strategy does not rely on a specified back EMF. No matter trapezoidal or sinusoidal back EMFs, the ZCP positions are the same and the mechanism is similar. Accordingly, this scheme has a good adaptive capacity and can overcome the parameter change.

C. Sampling Time-Delay Compensation

As mentioned above, the ZCP time intervals $t_{12} \ldots t_{61}$ are required in (24) and (29) to extract the ZCP angle errors. In this subsection, a strategy is introduced to obtain these ZCP time intervals with sampling time-delay compensation. As shown in Fig. 10, the ADC operates in synchronization with PWM generation. The terminal voltage is sampled at the middle point of PWM-ON duration. Actually, the zero-crossing event occurs between two successive sampling instants. For instance, in sector X, the ZCP of threshold u_x occurs between $v_x[n-1]$ and $v_x[n]$. Thus, when the ZCP is confirmed at the instant of $v_x[n]$, it leads to a sampling time-delay of Δt_x. As well, a time-delay Δt_y is introduced in ZCP detection of sector Y. Thus, the raw measurement result of ZCP time interval using CPU timer, denoted by

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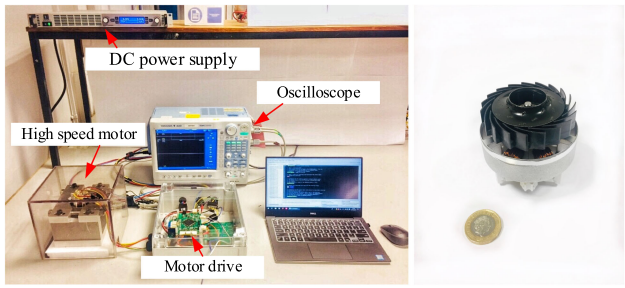


Fig. 11. Experimental test setup of high-speed motor drive. (a) Overall platform. (b) High-speed BLDC motor mechanically coupled with fan blades.

t'_{xy}, is not precise. Two time-delay terms Δt_x and Δt_y should be taken into consideration.

With the method of interpolation, the ZCP sampling time-delay in sector X can be calculated as

$$\Delta t_x = \frac{v_x\,[\mathrm{n}] - u_x}{v_x\,[\mathrm{n}] - v_x\,[\mathrm{n}-1]} \cdot T_{\mathrm{PWM}}. \tag{33}$$

Then, Δt_y can be calculated similarly. With these two time-delay signals Δt_x and Δt_y, the time interval of two successive ZCP events can be calculated as

$$t_{xy} = t'_{xy} + \Delta t_x - \Delta t_y. \tag{34}$$

By means of (34), the result t_{xy} with higher precision can be calculated. Six latest ZCP time interval results $(t_{12} \ldots t_{61})$ can be obtained and be adopted in (24) and (29).

After the ZCP is detected, the commutation instant that is 30° after each ZCP can be determined. For the conventional method, the time span of 30° is usually derived from half of ZCP time interval, i.e., $\frac{1}{2}t_{xy}$. When considering the sampling time-delay, we can implement the 30° delay with higher precision. When the zero-crossing event is detected at $v_y\,[\mathrm{n}]$, the commutation instant is actually $\frac{1}{2}t_{xy} - \Delta t_y$ after the sampling point, not $\frac{1}{2}t_{xy}$. Thus, the timer B can be set as

$$t_{\mathrm{timer}} = \tfrac{1}{2}t_{xy} - \Delta t_y. \tag{35}$$

V. Experimental Evaluation

A. Experimental Setup

The proposed adaptive threshold correction strategy is verified at the high-speed motor platform in Fig. 11. The high-speed BLDC motor is mechanically coupled with fan blades to produce the load torque. The DSP chip TMS320F28335 is adopted to execute all sensorless control algorithm. The PWM switching frequency and sampling frequency are both 40 kHz and the dc-link voltage is 15 V. The main parameters of the tested high-speed motor are given in Table II in Appendix B, in which we can see that the resistance and inductance values are asymmetric. In addition, to simulate the resistance tolerance of RVD, three different resistance ratios are used in three phases as shown in Table III in Appendix B. The equivalent gains K_{R} are 110%, 100%, and 90% in phases A, B, and C, respectively. It should be noted that no capacitors are utilized in the back EMF measurement circuits to avoid the phase delay of LPFs.

B. Experimental Results

Fig. 12 shows the comparison results with and without the proposed adaptive threshold correction scheme at 20 000 r/min. From top to bottom, the estimated sector number, sector-ZCP interval, estimated commutation instants and three-phase currents are given, respectively. The period is 3 ms and thus the motor commutes twice in each millisecond. It is convenient to judge the commutation accuracy by observing the estimated commutation instant signal in the grids.

Without correction as shown in Fig. 12(a), the sector-ZCP interval $(\hat{\theta}_{12} \ldots \hat{\theta}_{61})$ varies from 50° to 75°. Due to the machine parameter asymmetry and the resistance tolerance of RVD, the sector-ZCP interval cannot remain at 60°, but fluctuates within a wide range. Also, it is obvious that the commutation instants occur with inconsistent intervals. In this case, the current fluctuates drastically, the peak–peak value reaches 7.44 A.

When the vertical correction is enabled in Fig. 12(b), the sector-ZCP interval $(\hat{\theta}_{12} \cdots \hat{\theta}_{61})$ becomes closer to 60°, ranging from 55° to 65°. The commutation is still executed with inconsistent intervals, as the commutation instant signal is not aligned with the grids. The fluctuation can be observed in three-phase currents, with the peak–peak value 6.52 A. The vertical shift effect caused by the tolerance of RVD has been eliminated, so the remaining inconsistency is mainly caused by machine parameter asymmetry.

When both vertical and horizontal corrections are enabled in Fig. 12(c), the sector-ZCP interval $(\hat{\theta}_{12} \cdots \hat{\theta}_{61})$ converges to 60°. The motor commutes with consistent intervals and the commutation instant signal is aligned with the grids. The phase currents maintain almost identical in each sector. The current peak–peak value is 6.02 A, reduced by 19% compared to Fig. 12(a). Accordingly, both the vertical and horizontal envelope shift effects are eliminated with the proposed correction method.

Fig. 13 shows the comparison results with and without the proposed adaptive threshold correction strategy at 54 000 r/min (900 Hz). The PWM duty reaches 100% and thus it is the maximum speed. Without correction as shown in Fig. 13(a), the sector-ZCP interval cannot stay at 60°. The currents fluctuate with peak–peak value 26.2 A. With only vertical correction enabled, the sector-ZCP interval becomes closer to 60°, but still cannot remain at 60°. When both vertical and horizontal corrections are enabled, the sector-ZCP interval converges to 60°. Three-phase currents maintain almost identical and the phase current peak–peak value is reduced to 22.2 A. Therefore, the proposed adaptive threshold correction scheme is effective at rated speed.

Fig. 14 shows the convergence process of the zero-crossing thresholds at 18 000 r/min. From top to bottom, the sector-ZCP interval $(\hat{\theta}_{12} \cdots \hat{\theta}_{61})$, phase-ZCP-interval $(\hat{\theta}_{63} \cdots \hat{\theta}_{41})$ and the zero-crossing thresholds for each corresponding phase are presented. The vertical correction is enabled at 2 s. Then the phase-ZCP-interval gradually converges to 180°. As can be seen here, although the three phase-ZCP-intervals are all corrected to 180°,

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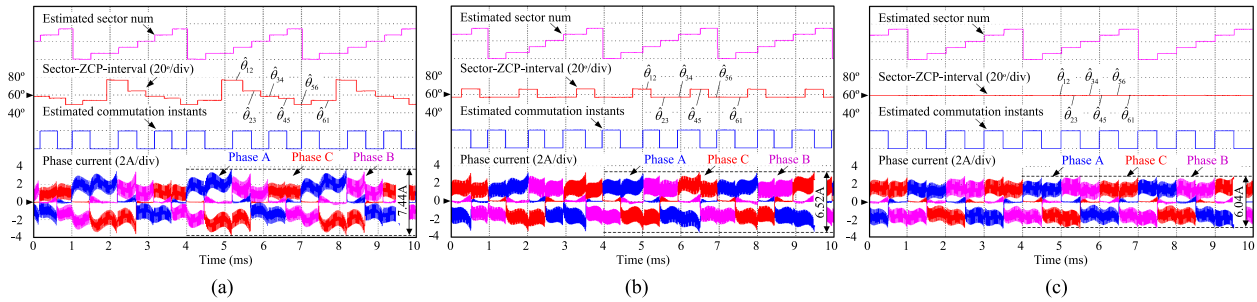


Fig. 12. Comparison results with and without the proposed adaptive threshold correction scheme at 20 000 r/min (333 Hz). (a) Without correction. (b) With only vertical correction. (c) With both vertical and horizontal corrections.

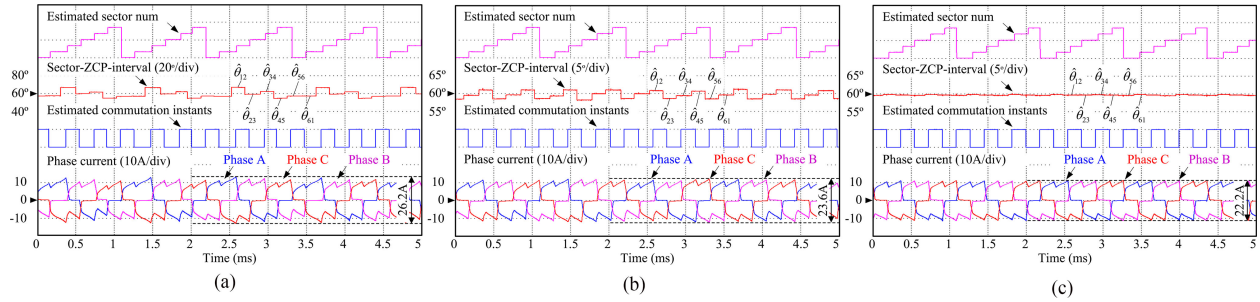


Fig. 13. Comparison results with and without the proposed adaptive threshold correction scheme at 54 000 r/min (900 Hz). (a) Without correction. (b) With only vertical correction. (c) With both vertical and horizontal corrections.

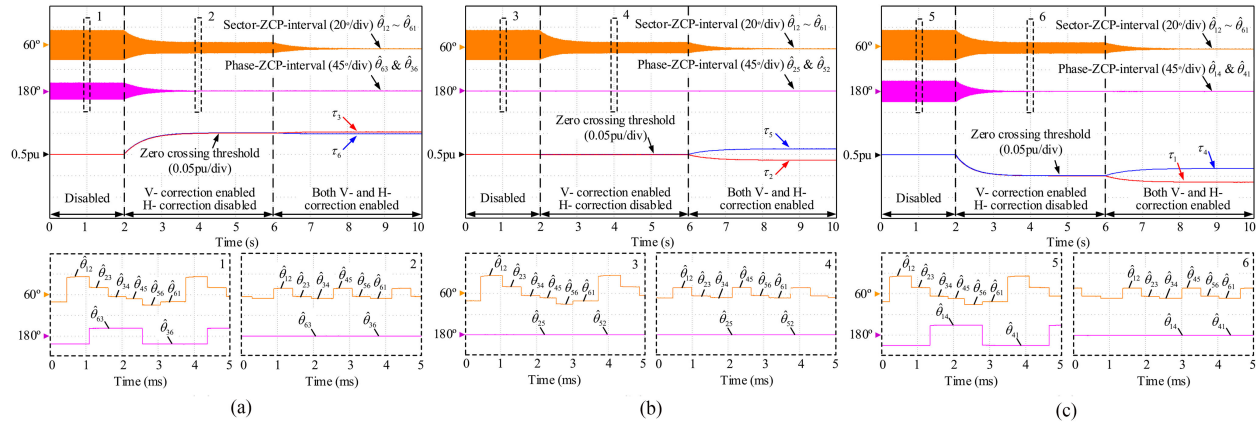


Fig. 14. Convergence process of zero-crossing thresholds for (a) phase A, (b) phase B, and (c) phase C.

the sector-ZCP interval still deviates from 60°. The vertical shift caused by RVD tolerance is eliminated. The remaining deviation is caused by the machine parameter asymmetry. For phases A and C, the zero-crossing thresholds converges to 0.55 and 0.45 p.u., respectively, which are in accord with their corresponding resistance ratio of RVD. At 6 s, the horizontal correction is enabled. Then the sector-ZCP interval converges to 60° by adjusting the six zero-crossing thresholds. Finally, with the vertical and horizontal corrections enabled, each phase-ZCP-interval converges to 180° and sector-ZCP interval converges to 60°. The results verify that the proposed strategy has a strong adaptive capacity for parameter change, reflecting its engineering application value.

Fig. 15 shows the sensorless operation at acceleration and deceleration processes. The motor accelerates from 18 000 to 54 000 r/min within only 3 s, and then decelerates back to 18 000 r/min with the same slope. When no correction is utilized in Fig. 15(a), the sector-ZCP interval fluctuates from 50° to 75° at 18 000 r/min. In the zoomed window, it is obvious that the speed begins to oscillate and the phase currents fluctuate drastically. It eventually triggers over-current protection and fails during acceleration. With only vertical correction enabled in

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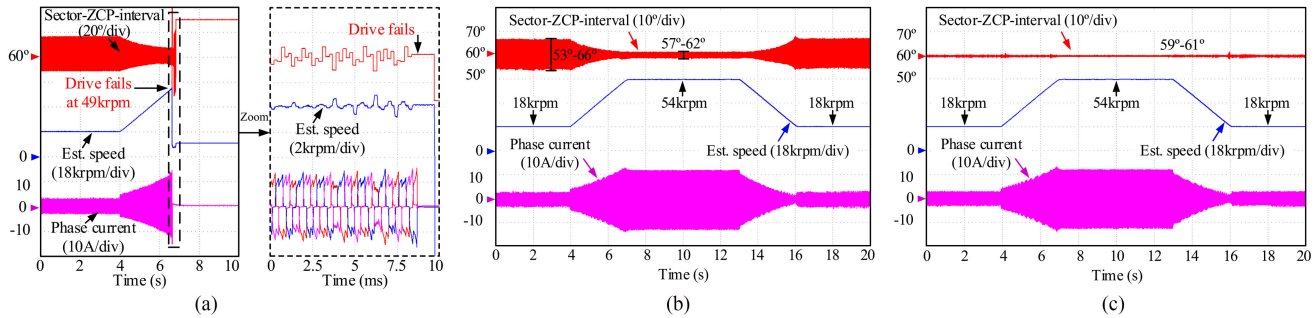


Fig. 15. Experimental comparison of sector ZCP angle interval at acceleration and deceleration processes. (a) Without correction. (b) With only vertical correction. (c) With both vertical and horizontal corrections.

Fig. 15(b), the sector-ZCP interval fluctuation range is 53°–66° at 18 000 r/min, then shrinks to 57°–62° at 54 000 r/min. Finally, when both vertical and horizontal corrections are enabled in Fig. 15(c), the fluctuation range maintains in 59°–61° during the whole operation. It can be seen that with the proposed adaptive threshold correction technique, the sector-ZCP interval can be accurately kept at 60° in a wide speed range.

VI. Conclusion

This article investigated the influence of asymmetric machine parameters and the resistance tolerance of back EMF measurement circuit on ZCP detection-based sensorless control of high-speed BLDC motor. First, these two nonideal factors were analyzed theoretically. It showed that when machine parameters were asymmetric, the back EMF envelope would deviate horizontally, whereas inaccurate back EMF sampling caused by the resistance tolerance of measurement circuit would cause a vertical deviation of envelopes. These two nonideal factors would lead to commutation errors and deteriorate the control performance. Based on the theoretical analysis, an adaptive threshold correction strategy was proposed to suppress the commutation errors. Besides, this strategy does not rely on the parameters of the machine or measurement circuits, which makes it suitable for batch production. Experimental results demonstrated that the proposed method can eliminate the commutation errors caused by the two nonideal factors and improved the performance of the sensorless high-speed BLDC drive. Furthermore, asymmetric mutual inductances that were not analyzed in this article could be a key issue for future investigations.

Appendix A

In order to calculate $\mathcal{L}^{-1}[V_{\mathrm{BG}}(s)]$, it should get every term in (5). Around the ZCP instant of sector II, $\theta = \omega t + 2\pi/3$, it yields

$$e_{\mathrm{A}}(t) - e_{\mathrm{C}}(t) = \sqrt{3}\lambda\omega\cos(\omega t). \tag{A1}$$

Its Laplace transformation is

$$E_{\mathrm{A}}(s) - E_{\mathrm{C}}(s) = \frac{\sqrt{3}\lambda\omega s}{s^2+\omega^2}. \tag{A2}$$

In PWM-ON period of sector II, $v_{\mathrm{AG}}(t)$ is a step of V_{DC} and $v_{\mathrm{CG}}(t)$ is zero, thus $V_{\mathrm{AG}}(s) = \frac{1}{s}V_{\mathrm{DC}}$ and $V_{\mathrm{CG}}(s) = 0$. Substitute all these terms into (5), then $\Delta V_{\mathrm{H}}(s)$ in sector II can be expressed as

$$\Delta V_{\mathrm{H2}}(s) = \left[\frac{1}{s}V_{\mathrm{DC}} - \frac{\sqrt{3}\lambda\omega s}{s^2+\omega^2}\right]\frac{\Delta R_{\mathrm{AC}} + s\Delta L_{\mathrm{AC}}}{\Sigma R_{\mathrm{AC}} + s\Sigma L_{\mathrm{AC}}}. \tag{A3}$$

Although the inverse Laplace term $\mathcal{L}^{-1}[\Delta V_{\mathrm{H}}(s)]$ is difficult to calculate, its initial value can be easily estimated as

$$\begin{aligned}\mathcal{L}^{-1}[\Delta V_{\mathrm{H2}}(s)] &\approx \lim_{t\to 0}\Delta v_{\mathrm{H2}}(t) = \lim_{s\to\infty} s\cdot\Delta V_{\mathrm{H2}}(s)\\ &= \lim_{s\to\infty} s\cdot\left[\tfrac{1}{s}V_{\mathrm{DC}} - \tfrac{\sqrt{3}\lambda\omega s}{s^2+\omega^2}\right]\tfrac{\Delta R_{\mathrm{AC}}+s\Delta L_{\mathrm{AC}}}{\Sigma R_{\mathrm{AC}}+s\Sigma L_{\mathrm{AC}}}\\ &= \left(V_{\mathrm{DC}} - \sqrt{3}\lambda\omega\right)\tfrac{L_{\mathrm{A}}-L_{\mathrm{C}}}{L_{\mathrm{A}}+L_{\mathrm{C}}}.\end{aligned} \tag{A4}$$

The voltage shift $\Delta v_{\mathrm{H2}}(t)$in sector II will stay around the initial value in short duration within a PWM period, since this short duration is much less than the fundamental period of motor drive $2\pi/\omega$ and the RL time constant L/R.

In a similar way, around the ZCP instant of sector V $\theta = \omega t + 5\pi/3$. Besides, $v_{\mathrm{AG}}(t)$ is zero and $v_{\mathrm{CG}}(t)$ is a step of V_{DC}, thus $V_{\mathrm{AG}}(s) = 0$ and $V_{\mathrm{CG}}(s) = \frac{1}{s}V_{\mathrm{DC}}$. Then, it yields

$$\Delta V_{\mathrm{H5}}(s) = \left[-\frac{1}{s}V_{\mathrm{DC}} + \frac{\sqrt{3}\lambda\omega s}{s^2+\omega^2}\right]\frac{\Delta R_{\mathrm{AC}} + s\Delta L_{\mathrm{AC}}}{\Sigma R_{\mathrm{AC}} + s\Sigma L_{\mathrm{AC}}}. \tag{A5}$$

Its initial value can be calculated as

$$\begin{aligned}\mathcal{L}^{-1}[\Delta V_{\mathrm{H5}}(s)] &\approx \lim_{t\to 0}\Delta v_{\mathrm{H5}}(t) = \lim_{s\to\infty} s\cdot\Delta V_{\mathrm{H5}}(s)\\ &= \left(-V_{\mathrm{DC}} + \sqrt{3}\lambda\omega\right)\tfrac{L_{\mathrm{A}}-L_{\mathrm{C}}}{L_{\mathrm{A}}+L_{\mathrm{C}}}\\ &= \left(V_{\mathrm{DC}} - \sqrt{3}\lambda\omega\right)\tfrac{L_{\mathrm{C}}-L_{\mathrm{A}}}{L_{\mathrm{C}}+L_{\mathrm{A}}}.\end{aligned} \tag{A6}$$

Appendix B

TABLE II
HIGH-SPEED BLDC MOTOR PARAMETERS

Parameter	Value
DC voltage	15 V
Rated current	14 A
Rated speed	54 000 r/min
Flux linkage	0.00127 V·s/rad
Pole pairs	1
Resistance (mΩ)	R_{A}=38.1 R_{B}=37.5 R_{C}=43.5
Inductance (uH)	L_{A}=27.5 L_{B}=27.2 L_{C}=28.6

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TABLE III
RESISTANCES OF RVD IN THREE PHASES

Phase	R1	R2	Resistance ratio	K_R
A	3.24 kΩ	11.5 kΩ	0.22	110%
B	3 kΩ	12 kΩ	0.2	100%
C	2.8 kΩ	12.7 kΩ	0.18	90%

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